

Note*: Please review the course textbook AIMA 4th Edition Chapter 3, Canvas Search Modules, and Assignment 1 before attempting this Challenge Question.

Glossary

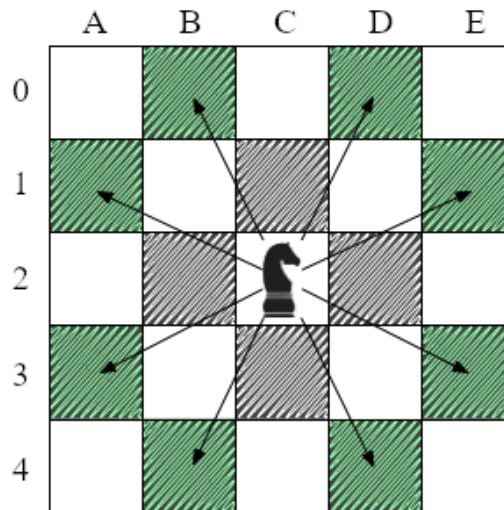


Figure 1: Valid moves for the Knight in Chess

- **Knight** - Symbolized by a horse head. It either moves up or down one square vertically and over two squares horizontally OR up or down two squares vertically and over one square horizontally. This movement can be remembered as an “L-shape” because it looks like a capital “L”. Note in Figure 1, the knight in square C2 has 8 valid moves and the valid moves are to positions D0, E1, E3, D4, B4, A3, A1, B0.

Introduction

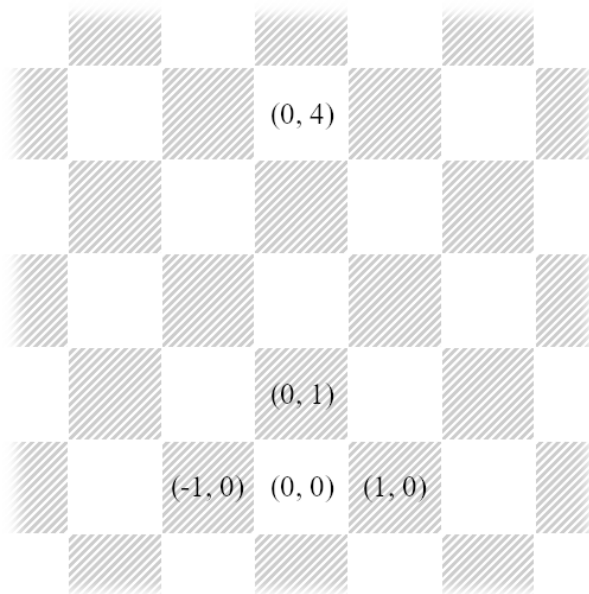


Figure 2: Infinite Chess Board

Imagine a knight located at the center of an infinite chess board (one that has no boundaries and expands infinitely in all directions), whose starting position we will denote to be $(0, 0)$ as seen in Figure 2. We may label all squares on the chess board using (x, y) notation relative to the starting square of $(0, 0)$ as seen in Figure 2. We'd like to move our knight to the square to goal position $(0, 4)$ which we will denote as position t , while incurring minimum cost.

Cost (incurred on every move) is calculated by the number of moves made to reach the current position. For example a **move sequence** of $\{(0, 0) \rightarrow (1, 2), (1, 2) \rightarrow (2, 4), (2, 4) \rightarrow (3, 6)\}$ has a cost of 3 because the knight was moved 3 times. We'd like to minimize this cost and the minimum possible cost (optimal) to reach the goal position is 2.

Clarifications:

1. You are allowed to visit the same square more than once.
2. Candidate move sequences with equivalent priority are removed from the search priority queue in First-In-First-Out (FIFO) order.
3. Candidate move sequences generated after exploring an action are sorted by their ϕ value before adding to the priority queue. ϕ is denoted as the angle (in degrees) between North (up) and the target position being moved to and takes on values in the range $0 \leq \phi < 360$. For example, if an action is to move from $(0, 1)$ to $(1, 3)$, it would have a ϕ value of 26.565° . Another example, if an action is to move from $(0, 1)$ to $(-1, -1)$, it would have a ϕ value of 206.565° . (This mainly affects FIFO behavior)
4. All questions in this Challenge Question refer to the infinite chess board in Figure 2 with a starting position of $(0, 0)$ and a goal of $(0, 4)$.

Q1 [1pt]

What is the **maximum branching factor** of the search tree?

Answer as a whole number.

Q2 [1pt]

Which of the following search techniques is guaranteed to find a solution (solution does not have to have minimum cost) in this problem? Assume infinite time is available. **Select all that apply.**

Multiple select.

- (a) Depth-First Search
- (b) Breadth-First Search
- (c) Iterative-Deepening Depth-First Search

Q3 [1pt]

Using the A* algorithm, which of the following heuristics will find an optimal solution (one that has the minimum possible cost of 2) in this problem? (Recall that $g(n)$ refers to the current cost, that $h(n)$ is the heuristic, and A* uses $g(n)+h(n)$ to determine which move sequence to explore next). **Select all that apply.**

Hint: These are not your conventional heuristics and so some theorems we take for granted might not hold true. When in doubt, try it out by hand!

Multiple select.

- (a) $h_1(n) = C$, where C is a real number constant ≤ 2
- (b) $h_2(n) = C - g(n)$, where C is a real number constant ≤ 2
- (c) $h_3(n) = 2g(n)$
- (d) $h_4(n) = -2g(n)$

Q4 [1pt]

Using the A* algorithm, which of the following heuristics will result in A* behaving like Breadth-First Search in this problem? **Select all that apply.**

Hint: These are not your conventional heuristics and so some theorems we take for granted might not hold true. When in doubt, try it out by hand!

Multiple select.

- (a) $h_1(n) = C$, where C is a real number constant ≤ 2
- (b) $h_2(n) = C - g(n)$, where C is a real number constant ≤ 2
- (c) $h_3(n) = 2g(n)$
- (d) $h_4(n) = -2g(n)$

Q5 [1pt]

Using the A* algorithm, which of the following heuristics will result in A* behaving like Uniform-Cost Search in this problem? **Select all that apply.**

Hint: These are not your conventional heuristics and so some theorems we take for granted might not hold true. When in doubt, try it out by hand!

Multiple select.

- (a) $h_1(n) = C$, where C is a real number constant ≤ 2
- (b) $h_2(n) = C - g(n)$, where C is a real number constant ≤ 2
- (c) $h_3(n) = 2g(n)$
- (d) $h_4(n) = -2g(n)$

Q6 [2pts]

Which of the following heuristics are **consistent** in this problem? Recall that a heuristic $h(n)$ is consistent if, for every node n and every successor n' of n generated by an action a , we have: $h(n) \leq c(n, a, n') + h(n')$. (AIMA 4th Ed. Section 3.5.2) **Select all that apply.**

Hint: These are not your conventional heuristics and so some theorems we take for granted might not hold true. Can you justify your answers by using the mathematical definition of consistency? Note that consistency may not always imply admissibility!

Multiple select.

- (a) $h_1(n) = C$, where C is a real number constant ≤ 2
- (b) $h_2(n) = C - g(n)$, where C is a real number constant ≤ 2

Q7 [3pts]

Let the set of landmarks be: $L = (0, 8), (8, 0), (0, -8), (-8, 0)$. For a knight located at position $(4, -2)$, what will the value of the landmark heuristic, $h_L(n)$ be in this problem?

Recall $h_L(n) = \min_{L \in \text{Landmarks}} C^*(n, L) + C^*(L, \text{goal})$. (AIMA 4th Ed. Section 3.6.4)

Answer as a whole number.